

An Open Source Ecosystem For 5G And Telecom Networks

As communications providers prepare to deliver high-speed connectivity to support new services and use cases, the demand for low-latency, high-bandwidth, scalable networks continues to grow. This may lead to software-defined infrastructure and cloud native approaches to meet 5G requirements. Here's a look at what Linux Foundation (LF) is working on to create an open source ecosystem for the same

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Successful open source developments depend on complete life-cycles of the projects that the market will adopt and deploy. There is an exciting convergence in the networking industry around open source, and the energy is palpable. In early 2021, the Linux Foundation (LF) announced a collaboration with Facebook to be managed under a neutral governance framework.

Magma, an open source software platform, enables operators to build and augment modern and efficient mobile networks at scale. It features an access-agnostic mobile packet core, advanced network automation and management tools, and the ability to integrate with existing LTE networks. It has use cases across both virtual and container network functions (xNFs) including fixed wireless access, carrier Wi-Fi, private LTE and 5G, network expansion, and mobile broadband.

In all these major areas, the LF hosts over

700 projects that are either in a technology area or a market area where innovation can speed up ongoing work in India. The numbers are just staggering these days. Most of the Fortune 500 companies use our quarter of a million developers, contributing code to almost 500 plus projects. And then if you look at just the stats through LFX insight, 31 million lines of code were added between 1.1 million poll requests and 10,000 repositories.

Technology disruption

The open source journey has been quite significant in the past decade, starting with virtualisation, desegregation, and software defined networking to control plane databases. We have come a long way in the Kubernetes cloud native and are now moving to intent based.

While earlier focus was to develop projects in technology, building blocks for open source, in the next decade we are going to see all these open source projects working together for a common solution. The problem we are trying to solve this year is how to move code into production as quickly as possible.

While there are projects like Anuket, which is in LF networking, projects which are LFH can solve exactly this problem. They look at blueprints and how to comply with a spec, do open verification, submit the test results and do open interrupt, test-ing, training, and certification. So that's

Fig. 1: LF open source component projects for 5G

